

Supplementary Materials

Neuromorphic Sequential Arena: A Benchmark for Neuromorphic Temporal Processing

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Theoretical and Empirical Computation of Energy Cost

In this section, we detail the derivation of energy consumption as presented in Table 5. Following standard practices for assessing energy costs of Spiking Neural Networks (SNNs), we quantify the number of 32-bit floating-point Multiply-Accumulate (MAC) operations and Accumulate (AC) operations involved in a single inference.

First, based on the temporal dynamics of each model, we derive theoretical energy consumption formulas, which are provided in Table S1. In these formulas, E_{MAC} and E_{AC} represent the energy required for each MAC and AC operation on neuromorphic hardware, respectively. m , n , and k denote the input size, hidden size, and kernel size, respectively. h is the hidden dimension of the feedforward module in Spike-Driven Transformer. The variable Fr_i captures the spike frequency of a spike sequence i , with examples including Fr_{in} and Fr_{out} , which represent the sparsity of input and output spike sequences for a given layer. For models with additional architectural components, such as the Spiking TCN, Fr_{conv2} specifies the sparsity of input spikes in the second convolution layer. Similarly, in the Spike-Driven Transformer, Fr_q , Fr_k , Fr_v , Fr_{attn} , Fr_{fc1} , and Fr_{fc2} denote the sparsity of LIF neuron populations in the self-attention and feedforward modules. For GSU, Fr_y and Fr_w represent the sparsity of ternary outputs and weights, respectively.

Subsequently, to derive these spike frequency values required for calculating the empirical energy consumption of the evaluated models, we perform inference on well-trained spiking models using the AL task with a sequence length of 400. All networks are configured with hidden dimensions of [256, 256], and the second hidden layer is consistently utilized for comparisons across evaluated models. During inference, we record the average spike frequency per time step, sample, and neuron for all spiking neuron populations involved in the computation.

Finally, based on data derived from a 45 nm CMOS process [Horowitz, 2014], the energy per operation is set to $E_{AC} = 0.9$ pJ for AC operations and $E_{MAC} = 4.6$ pJ for MAC operations. All these data are substituted into the theoretical energy cost formulas to compute the empirical energy cost, as outlined in the main text.

Effectiveness Evaluation on Existing SNN Benchmarks

In this section, we apply the proposed Segregated Temporal Probe (STP) criterion to evaluate commonly used datasets in the neuromorphic community, shedding light on their effectiveness as benchmarks for assessing temporal processing capacities. For each benchmark, we follow standard configurations for dataset preprocessing, data augmentation, network design, and training setup.

We first examine static image recognition datasets, including MNIST [LeCun *et al.*, 1998], CIFAR10 [Krizhevsky and Hinton, 2009], and CIFAR100 [Krizhevsky and Hinton, 2009]. These datasets lack temporal dynamics, as each input sequence is generated by repeating a static image across the temporal dimension. This limitation is evidenced by our experimental results. As shown in Table S2, NoTD and SDBP achieve accuracy comparable to STBP, demonstrating that modeling temporal relationships is unnecessary for high performance on these datasets.

Next, we analyze event-based vision datasets, including N-MNIST [Orchard *et al.*, 2015], CIFAR10-DVS [Li *et al.*, 2017], DVS-Gesture [Amir *et al.*, 2017], DVS-SLR [Zhou *et al.*, 2024], GAIT-DAY-DVS [Wang *et al.*, 2022], and HAR-DVS [Wang *et al.*, 2024]. Despite the presence of event-driven dynamics, our results presented in Table S3 reveal that NoTD performs similarly to or even outperforms STBP, indicating that these datasets can be effectively addressed at the frame level without the establishment of temporal dependency.

Furthermore, we evaluate keyword spotting datasets, including GSC [Warden, 2018], SHD [Cramer *et al.*, 2020], and SSC [Cramer *et al.*, 2020]. The results demonstrated in Table S4 reveal that effective decisions for these datasets can often be made by integrating short-term temporal features, as indicated by the marginal performance gap between STDP and STBP and the moderate accuracy achieved by NoTD. These findings suggest that temporal propagation primarily contributes to the integration of short-term features during the forward pass, while backward temporal propagation has a negligible impact on establishing temporal dependencies. Consequently, these benchmarks are inadequate for effectively evaluating the temporal processing capacities of SNNs.

These findings underscore the limitations of existing benchmarks and the urgent need for a more effective neuromorphic temporal processing benchmark. NSA addresses these gaps by incorporating tasks that contain rich, multi-timescale temporal

Table S1: Theoretical and empirical energy costs of different architectures and neuron models.

Architecture	Neuron model	Theoretical energy cost	Empirical energy cost (nJ)
FFSNN	LIF	$(mnFr_{in} + nFr_{out})E_{AC} + nE_{MAC}$	4.52
	CE-LIF	$(mnFr_{in} + nFr_{out})E_{AC} + 3nE_{MAC}$	6.43
	LTC	$(mnFr_{in} + 2nFr_{out} + 2n)E_{AC} + (4nn + 3n)E_{MAC}$	1,210.23
	SPSN	$mnFr_{in}E_{AC} + nkE_{MAC}$	152.14
	PMSN	$mnFr_{in}E_{AC} + (2n \log_2(T) + 5n)E_{MAC}$	30.68
SRNN	LIF	$(mnFr_{in} + (nn + n)Fr_{out})E_{AC} + nE_{MAC}$	21.79
	CE-LIF	$(mnFr_{in} + (nn + n)Fr_{out})E_{AC} + 3nE_{MAC}$	10.96
	LTC	$(mnFr_{in} + (nn + 2n)Fr_{out} + 2n)E_{AC} + (4nn + 3n)E_{MAC}$	1,211.3
GSN	LIF	$(2mnFr_{in} + 2nnFr_{out})E_{AC} + 5nE_{MAC}$	30.48
Spiking TCN		$(kmnFr_{in} + knnFr_{conv2})E_{AC}$	19.85
Spike-Driven Transformer		$((3Fr_{in} + Fr_{attn})nn + (Fr_qFr_k + Fr_v)nT + (Fr_{fc1} + Fr_{fc2})nh)E_{AC}$	193.22
Binary S4D		$2nnFr_{out}E_{AC} + (2n \log_2(T) + 8n)E_{MAC}$	65.01
GSU		$(nn(Fr_y + Fr_w) + 2n)E_{AC} + (2n \log_2(T) + 6n)E_{MAC}$	28.36

dependencies, ensuring an effective and consistent assessment of SNNs' temporal processing capacities.

Table S2: Validating the effectiveness of static image recognition datasets in evaluating the temporal processing capacity of SNNs.

Method	MNIST Acc. (%) ↑	CIFAR10 Acc. (%) ↑	CIFAR100 Acc. (%) ↑
STBP	99.40	94.86	74.57
SDBP	99.27 (-0.13)	94.74 (-0.12)	74.35 (-0.22)
NoTD	99.18 (-0.22)	93.46 (-1.40)	73.28 (-1.29)

Table S3: Validating the effectiveness of event-based vision datasets in evaluating the temporal processing capacity of SNNs.

Method	N-MNIST Acc. (%) ↑	CIFAR10-DVS Acc. (%) ↑	DVS-Gesture Acc. (%) ↑	DVS-SLR Acc. (%) ↑	GAIT-DAY-DVS Acc. (%) ↑	HARDVS Acc. (%) ↑
STBP	99.49	78.50	95.14	76.57	93.52	48.64
SDBP	99.48 (-0.01)	79.00 (+0.50)	95.83 (+0.69)	75.92 (-0.65)	93.70 (+0.18)	48.01 (-0.63)
NoTD	99.09 (-0.40)	80.00 (+1.50)	94.44 (-0.70)	75.18 (-1.39)	93.47 (-0.05)	47.98 (-0.66)

Experimental Setups

Computing Infrastructure

All experiments are conducted on an Ubuntu 20.04.5 LTS server equipped with NVIDIA GeForce RTX 3090 GPUs (24GB), an Intel Xeon Platinum 8370C CPU, PyTorch 1.13.0, and CUDA 11.8.

Table S4: Validating the effectiveness of keyword spotting datasets in evaluating the temporal processing capacity of SNNs.

Method	GSC	SHD	SSC
	Acc. (%) $\uparrow$	Acc. (%) $\uparrow$	Acc. (%) $\uparrow$
STBP	92.91	86.48	67.13
SDBP	89.00 (-3.91)	85.07 (-1.41)	66.03 (-1.10)
NoTD	77.53 (-15.38)	68.51 (-17.97)	44.97 (-22.16)

Data Preprocessing

- **AL:** In this task, we utilize our proposed synthetic dataset AL. Each sample begins with a robot positioned at coordinate (0, 0), facing the positive y-axis. The robot then executes a randomly generated sequence of actions, with each action sampled from the set {‘turn left’, ‘turn right’, ‘go straight’, ‘stop’} according to the probabilities [0.05, 0.05, 0.45, 0.45], respectively. At the end of the sequence, the robot’s final x-coordinate determines the classification: $x \leq 0$ is “left”, otherwise “right”.
- **HAR:** The smartwatch gyroscope data (sampled at 20 Hz) from the WISDM dataset [Weiss, 2019] is segmented using a sliding window approach. Each window has a length of 200 time steps, with a hop length of 100 time steps. The data features angular velocity in three coordinates (X, Y, Z). To eliminate subject-specific bias and ensure consistency, data from all subjects is z-score normalized per coordinate, shuffled, and split into training and testing sets.
- **EEG-MI:** EEG motor imagery data from the OpenBMI dataset [Lee *et al.*, 2019] is preprocessed using a widely adopted EEG signal preprocessing pipeline [Hosseini *et al.*, 2020]. The raw signals are bandpass-filtered (0.5–80 Hz), down-sampled to 100 Hz, and cleaned using Independent Component Analysis (ICA) to remove artifacts. The data are then segmented to extract the principal EEG data relevant to the task. Subsequently, each channel is z-score normalized to ensure feature consistency. Finally, the samples are shuffled across subjects and split into training and testing sets to eliminate subject-specific biases and ensure effective model evaluation.
- **SSL:** We preprocess the raw audio signals (sampled at 16 kHz) provided by the SLoClas dataset [Qian *et al.*, 2021] using a sliding window approach with a window size of 500 and a hop size of 500. To simulate real-world application scenarios, directional background noise from 0°, 90°, 180°, and 270° provided in the dataset are randomly segmented and added to each sample at 0 SNR. Notably, raw time-domain signals are processed directly, without the application of any frequency spectral analysis.
- **ALR:** We preprocess the raw event data in the DVS-Lip dataset [Tan *et al.*, 2022] by converting asynchronous events into frame-based representations. Specifically, events are grouped into their nearest temporal bins, forming T temporal bins, where $T = 200$ in our experiments. To preserve polarity information, positive and negative events are mapped into two separate channels. For data augmentation, we apply a series of transformations to the training data, including center cropping and random cropping to achieve an 88×88 spatial resolution, horizontal flipping with a probability of 0.5, cutout, and random zoom-in/zoom-out. For the test data, only center cropping to 88×88 is applied.
- **AD:** All audio samples in the N-DNS dataset [Timcheck *et al.*, 2023] are processed at a sampling rate of 16 kHz and standardized to a fixed duration of 30 seconds. For audio clips shorter than 30 seconds, we concatenate additional speech segments from the same speaker, inserting a 0.2-second silence between utterances to maintain natural pauses. Noisy audio is generated by mixing clean speech with randomly selected noise samples at signal-to-noise ratios ranging from -5 dB to 20 dB, ensuring a diverse range of noise conditions. To simulate varying input loudness levels, we apply loudness normalization to each noisy sample, adjusting levels between -35 dB and -15 dB relative to full scale. For spectral analysis, the Short-Time Fourier Transform (STFT) is configured with a 32 ms window length (512 samples) and an 8 ms hop length (128 samples), using a Hanning window and 512-point Fast Fourier Transform to capture detailed frequency information.
- **ASR:** Experiments on the AISHELL dataset [Bu *et al.*, 2017] are conducted using the ESPnet [Watanabe *et al.*, 2018] toolkit, following its standard recipes for data preparation, model training, and evaluation. This ensures a consistent and reproducible experimental setup.

Task-Specific Configurations

In all experiments, the output layer of the SNNs is implemented as a non-spiking fully connected readout layer. For AL, HAR, and SSL tasks, the final network output is obtained by averaging the readout layer’s outputs across all time steps. For EEG-MI and ALR tasks, the network output is derived solely from the readout layer’s output at the final time step. For AD and ALR tasks, where labels vary across time steps, the network output at each time step is dynamically determined by the corresponding readout layer’s output.

For AL, HAR, EEG-MI, SSL, and ALR tasks, we use the AdamW optimizer with Cross-Entropy (CE) loss. In AD experiments, following [Hao *et al.*, 2024], we adopt a combination of magnitude loss and complex loss in the time-frequency domain

alongside the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) loss. For ASR experiments, the Connectionist Temporal Classification (CTC) loss is employed. Additionally, the triangle surrogate function [Deng *et al.*, 2022] is consistently applied across all experiments.

In **Benchmark Effectiveness Validation (Sec. 3.1)**, the sequence length of the AL task is adjusted to 200 thereby highlighting the performance gap among the three training algorithms in STP. All experiments in this section follow the hyperparameters and model configurations of the LIF-SFNN models listed in Tables S5 and S6.

In **Performance Benchmarking of SNN Models (Sec. 3.2)**, the default sequence length for the AL task is extended to 400. To ensure fair comparisons, all models are configured with an equivalent number of trainable parameters to the baseline SFNN with LIF neurons for each task. For AL, HAR, EEG-MI, and SSL tasks, the number of trainable parameters is set to approximately $0.1M$ for all models. For ALR, AD, and ASR tasks, the trainable parameters are approximately $9.5M$, $0.4M$, and $4.7M$, respectively. Detailed hyperparameters and model specifications for all tasks are provided in Tables S5 and S6. In terms of model-specific hyperparameters, β represents the threshold decay constant for CE-LIF. For SPSN and Spiking TCN models, k denotes the kernel size, which defines the receptive field over time for temporal feature extraction. In the Spike-Driven Transformer, $nhead$ specifies the number of attention heads in the multi-head attention mechanism. Additionally, wd is an abbreviation for weight decay, which is utilized to regularize spatial credit assignment in Spiking S4D and GSU models.

In **Efficiency Benchmarking of SNN Models (Sec. 3.3)**, we evaluate training speed and memory cost using sequence lengths of 200, 400, and 800 on the AL task, while setting the sequence length to 400 for energy efficiency evaluation. All SNN models are standardized with two hidden layers of 256 channels each, ensuring consistent hidden dimensions for fair comparisons. The batch size is fixed at 256, and other model-specific hyperparameters remain unchanged from prior experiments.

Table S5: Hyperparameter configurations and model specifications for AL, HAR, EEG-MI, and SSL tasks.

Task	Architecture	Neuron model	Batch size	Epochs	Hidden dimension	Learning rate	Grad clipping	Decay	Threshold	Surrogate gradient width	Others
AL	FFSNN	LIF	256	100	128-256-256	5.00E-03	1	1	0.5	0.4	$\beta=0.02$
		CE-LIF	256	100	128-152-152	3.00E-03	1	0.5	0.5	0.4	
		LTC	256	100	96-128	1.00E-03	1	-	0.5	0.4	
		SPSN	256	100	128-256-256	5.00E-03	1	-	0.5	0.4	
		PMSN	256	100	64-256-256	2.00E-02	1	-	0.5	1	
	SRNN	LIF	256	100	128-176-176	1.00E-03	1	0.5	0.5	0.4	$\beta=0.03$
		CE-LIF	256	100	120-120-120	1.00E-03	1	0.3	0.5	0.4	
		LTC	256	100	96-112	2.00E-03	0.25	-	0.5	0.4	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	256	100	64-112-112	3.00E-03	1	-	0.5	0.6	$k=7$ $nhead=8$ $wd=3.00E-03$ $wd=3.00E-03$
			256	100	36*6	5.00E-03	1	0.5	0.5	1	
			256	100	64-64	5.00E-04	1	0.5	0.5	0.6	
			256	100	128-128	3.00E-03	1	-	0.5	0.6	
			256	100	152-152	5.00E-03	1	-	0.5	0.6	
HAR	FFSNN	LIF	256	100	128-256-256	3.00E-03	1	0.9	0.5	0.6	$\beta=0.15$
		CE-LIF	256	100	128-192-192	3.00E-03	1	0.3	0.5	0.6	
		LTC	256	100	96-128	3.00E-03	1	-	0.5	0.6	
		SPSN	256	100	128-256-256	3.00E-03	1	-	0.5	0.6	
		PMSN	256	100	64-256-256	5.00E-03	1	-	0.5	1	
	SRNN	LIF	256	100	128-176-176	1.50E-03	0	0.1	0.5	0.6	$\beta=0.1$
		CE-LIF	256	100	128-128-160	3.00E-03	1	0.3	0.5	0.6	
		LTC	256	100	96-112	3.00E-03	1	-	0.5	0.6	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	256	100	64-112-112	5.00E-03	1	-	0.5	0.6	$k=16$ $nhead=1$ $wd=3.00E-03$ $wd=3.00E-03$
			256	100	76	5.00E-03	1	0.5	0.5	1	
			256	100	64-64	3.00E-03	1	0.5	0.6	0.8	
			256	100	128-128	1.00E-02	1	-	0.5	0.6	
			256	100	152-152	1.00E-02	1	-	0.5	0.6	
EEG-MI	FFSNN	LIF	128	100	128-256-256	1.00E-03	1	1	0.5	0.6	$\beta=0.02$
		CE-LIF	128	100	128-128-128	1.00E-03	1	0.3	0.5	0.6	
		LTC	128	100	96-112	1.00E-03	0	-	0.5	0.6	
		SPSN	128	100	128-256-256	3.00E-03	1	-	0.5	0.6	
		PMSN	128	100	64-256-256	1.00E-02	1	-	0.5	1	
	SRNN	LIF	128	100	128-176-176	2.00E-04	1	1	0.5	0.6	$\beta=0.02$
		CE-LIF	128	100	64-128-128	2.00E-04	1	0.2	0.5	0.6	
		LTC	128	100	96-104	1.00E-03	0	-	0.5	0.6	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	128	100	64-112-112	5.00E-03	1	-	0.5	0.6	$k=7$ $nhead=8$ $wd=5.00E-03$ $wd=5.00E-03$
			128	100	28*8	3.00E-03	1	0.5	0.5	1	
			128	100	64-64	5.00E-04	1	0.5	0.5	0.6	
			128	100	128-128	1.00E-02	1	-	0.5	0.6	
			128	100	152-152	1.00E-02	1	-	0.5	0.6	
SSL	FFSNN	LIF	64	100	128-256-256	5.00E-03	1	0.6	0.5	0.6	$\beta=0.05$
		CE-LIF	64	100	64-128-128	5.00E-03	1	0.3	0.5	0.6	
		LTC	64	100	96-112	3.00E-03	1	-	0.5	0.6	
		SPSN	64	100	128-256-256	3.00E-03	1	-	0.5	0.6	
		PMSN	64	100	64-256-256	3.00E-03	1	-	0.5	1	
	SRNN	LIF	64	100	128-168-168	5.00E-04	1	0.5	0.5	0.4	$\beta=0.02$
		CE-LIF	64	100	96-108-108	1.00E-03	1	0.3	0.5	0.6	
		LTC	64	100	96-104	1.00E-03	0.5	-	0.5	0.6	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	64	100	96-112-112	1.00E-03	0.15	-	0.5	0.4	$k=3$ $nhead=4$ $wd=5.00E-03$ $wd=5.00E-03$
			64	100	46*8	3.00E-03	1	0.5	0.5	1	
			64	100	64-64	5.00E-04	1	0.5	0.5	0.6	
			64	100	128-128	1.00E-02	1	-	0.5	0.6	
			64	100	152-152	1.00E-02	1	-	0.5	0.6	

Table S6: Hyperparameter configurations and model specifications for ALR, AD, and ASR tasks.

Task	Architecture	Neuron model	Batch size	Epochs	Hidden dimension	Learning rate	Grad clipping	Decay	Threshold	Surrogate gradient width	Others
ALR	FFSNN	LIF	256	100	512*6	3.00E-03	0	0.95	0.8	1	$\beta=0.02$
		CE-LIF	256	100	512*6	3.00E-03	1	0.3	0.5	1	
		LTC	256	100	512*6	3.00E-03	0	-	0.8	1	
		SPSN	256	100	512*6	3.00E-03	0	-	0.8	1	
		PMSN	256	100	512*6	3.00E-03	0	-	0.8	1	
	SRNN	LIF	256	100	460*6	5.00E-04	0	0.95	0.8	1	$\beta=0.02$
		CE-LIF	256	100	460*6	3.00E-03	1	0.3	0.5	1	
		LTC	256	100	340*6	5.00E-04	0	-	0.8	1	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	256	100	256*6	5.00E-04	0	-	0.1	1	$k=7$ $nhead=4$ $wd=5.00E-04$ $wd=5.00E-04$
			256	100	75*6	3.00E-03	0	0.5	0.5	1	
			256	100	272*6	5.00E-04	0	0.5	0.8	1	
			256	100	435*6	3.00E-03	0	-	0.8	1	
			256	100	485*6	3.00E-03	0	-	0.8	1	
AD	FFSNN	LIF	64	50	320*2/224*2	1.00E-03	10	0.8	0.5	1	$k=32$
		LTC	64	50	140*2/85*2	1.00E-03	10	-	0.5	1	
		SPSN	64	50	320*2/224*2	1.00E-03	10	-	0.5	1	
		PMSN	64	50	320*2/224*2	5.00E-04	10	-	0.5	1	
	SRNN	LIF	64	50	280*2/160*2	1.00E-03	0	0.5	0.5	1	
		LTC	64	50	136*2/80*2	1.00E-03	10	-	1.0	1	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	64	50	155*2/100*2	1.00E-03	10	-	0.5	1	$k=3$ $nhead=4$
			64	50	125*2/75*2	1.00E-03	10	0.5	0.5	1	
			64	50	84*2/56*2	5.00E-04	10	0.5	0.5	1	
			64	50	170*2/100*2	1.00E-03	10	-	0.5	1	
			64	50	195*2/130*2	1.00E-03	10	-	0.5	1	
ASR	FFSNN	LIF	360	20	1024*3	2	0	0.95	0.8	1	$\beta=0.2$
		CE-LIF	360	20	1024*3	2	0	0.5	0.5	0.6	
		LTC	360	20	512*3	2	0	-	0.8	1	
		SPSN	360	20	1024*3	2	0	-	0.5	1	
		PMSN	360	20	1024*3	1	0	-	0.5	1	
	SRNN	LIF	360	20	820*3	2	0	0.3	0.5	1	$\beta=0.2$
		CE-LIF	360	20	770*3	2	0	0.3	0.5	0.6	
		LTC	360	20	480*3	2	0	-	1.0	0.8	
	GSN Spiking TCN Spike-Driven Transformer Binary S4D GSU	LIF	360	20	512*3	2	0	-	0.8	0.6	$k=7$ $nhead=4$
			360	20	128*3	2	0	0.5	0.5	1	
			360	20	328*3	1	0	0.5	0.3	1	
			360	20	680*3	2	0	-	0.5	0.6	
			360	20	850*3	2	0	-	0.5	0.6	